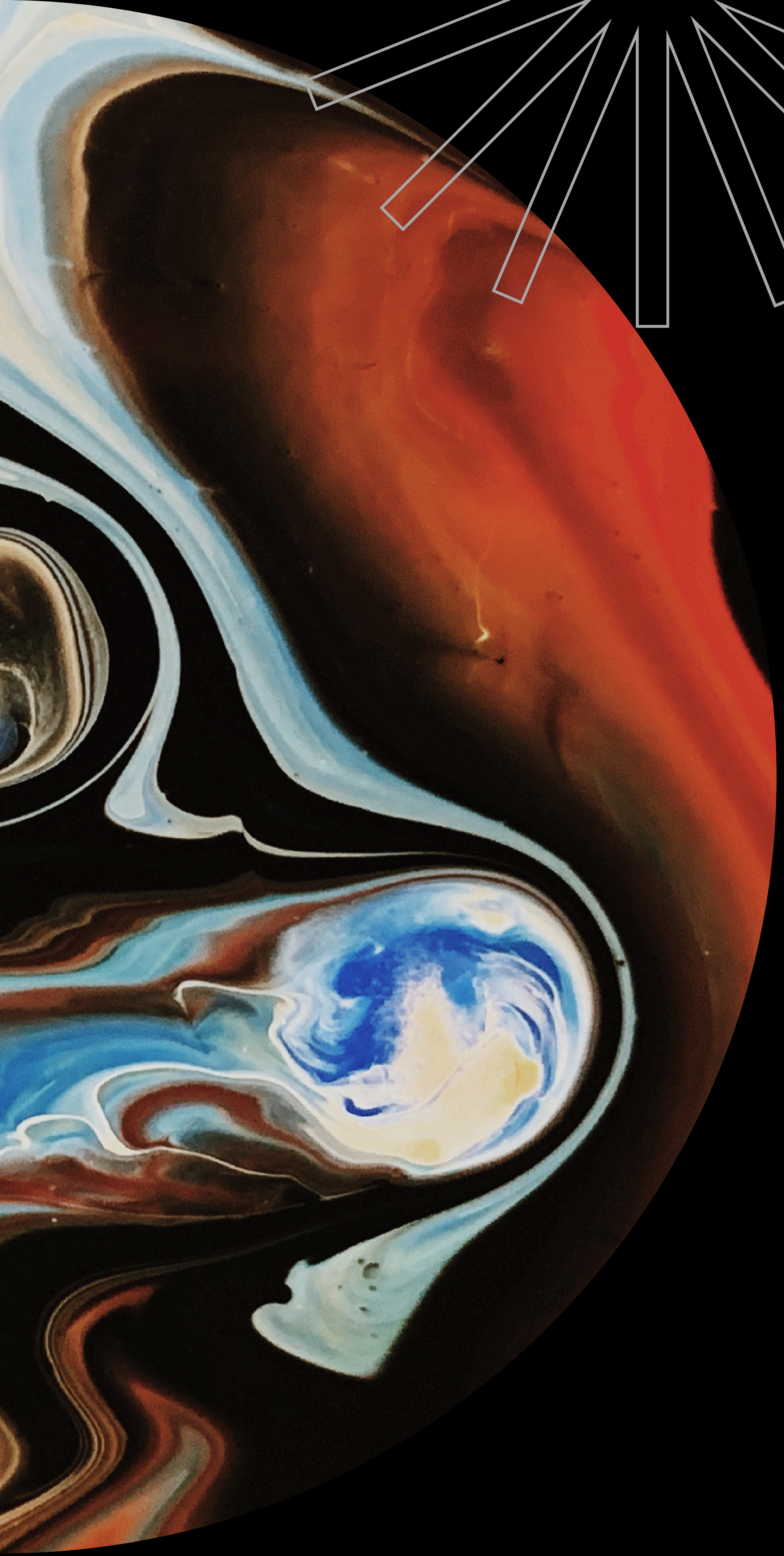
PORTFOLIO

DEVELOPER, DESIGNER



TANUSH THAKRAN

AI-focused developer building applied machine learning and LLM systems, from retrieval-augmented generation pipelines to NLP-driven products. Currently pursuing an Integrated B.Sc. in Computer Science at BITS Pilani. Comfortable across the full stack — Python for AI/data workloads and MERN + Flutter for shipping intelligent, user-facing applications. Seeking AI/ML engineering and applied-AI roles where I can turn models into production systems

INTRODUCTION

KEY FEATURES

- Document Ingestion & Processing – Upload and process documents into searchable text chunks for efficient retrieval.
- Semantic Vector Search – Uses embeddings and vector similarity search to retrieve the most relevant context for every query.
- Context-Aware AI Responses – Generates accurate answers grounded in retrieved document content, minimizing hallucinations.
- Interactive Chat Interface – Ask natural language questions and receive conversational, source-aware responses.
- Scalable RAG Pipeline – Modular architecture that supports adding new documents and knowledge bases without retraining the language model.

RAG BASED AI

RAG Based AI is an intelligent document question-answering system that leverages Retrieval-Augmented Generation (RAG) to provide accurate, context-aware responses from custom knowledge sources. Instead of relying solely on an LLM's pre-trained knowledge, the application retrieves the most relevant document chunks using semantic search and generates grounded answers, reducing hallucinations and improving factual accuracy. The system is designed to efficiently process user-provided documents, making it suitable for knowledge management, research assistance, and enterprise document search. RAG combines retrieval with generation to improve the quality and freshness of AI responses.

GitHub Repo Link : https://github.com/Tanush1206/rag_based_ai

KEY FEATURES

- Developed an intelligent document analysis system that processes legal contracts, agreements, and policies to extract key information and simplify complex terminology.
- Implemented Natural Language Processing (NLP) models to translate legal language into plain, user-friendly text while preserving the original meaning.
- Designed a summary generation feature that highlights important clauses, obligations, and risks in a structured and readable format.
- Built a clean and intuitive user interface allowing users to upload legal documents and receive simplified outputs within seconds.
- Enabled support for multiple document formats such as PDF and Word files.
- Ensured accuracy and clarity by fine-tuning AI models using real-world legal samples.

All The Way Transportation System (ATW)

Built an AI-driven platform designed to help non-legal users understand complex legal documents by automatically converting legal jargon into clear, concise, and easy-to-read language. The solution empowers individuals, startups, and small businesses to make faster, more informed decisions without requiring deep legal expertise.

GitHub Repo Link :
FrontEnd - https://github.com/NeuralSynth/atw-frontend
BackEnd - https://github.com/NeuralSynth/atw_backend

KEY FEATURES

- Natural Language Processing (NLP): Leveraged AI/LLMs to analyze legal contracts, agreements, and policies, breaking down technical terms into simplified explanations.
- Summarization & Highlighting: Automatically generates key takeaways, obligations, and risks, helping users quickly grasp critical information.
- Interactive Q&A Agent: Users can ask questions about specific clauses and receive AI-powered contextual answers.
- Comprehension Boost: Reduced average reading and understanding time by ~30%, improving user efficiency.
- User-Centric Design: Clean, intuitive UI tailored for individuals without a legal background.

PactPal

Built an AI-driven platform that helps non-legal users understand complex legal documents by automatically simplifying legal jargon into clear, concise, and easy-to-read text. The solution empowers individuals and businesses to make faster and more informed decisions without requiring deep legal expertise.

GitHub Repo Link :
FrontEnd - https://github.com/Tanush1206/PactPal-Frontend
BackEnd - https://github.com/Tanush1206/PactPal-Backend

Live Link :
https://pactpal-frontend.onrender.com/

CLIQUE

Modern full-stack event & house-competition platform for Scaler students (Google OAuth, real-time leaderboards).

CLIQUE is a campus event management system built exclusively for Scaler (@sst.scaler.com) students. It supports creating and running fests, hackathons and cultural events while tracking house points and live leaderboards with secure Google OAuth authentication and role-based access.

Github Repo Link :
https://github.com/Tanush1206/CLIQUE

Live Link :
https://frontend-clique-1.onrender.com/

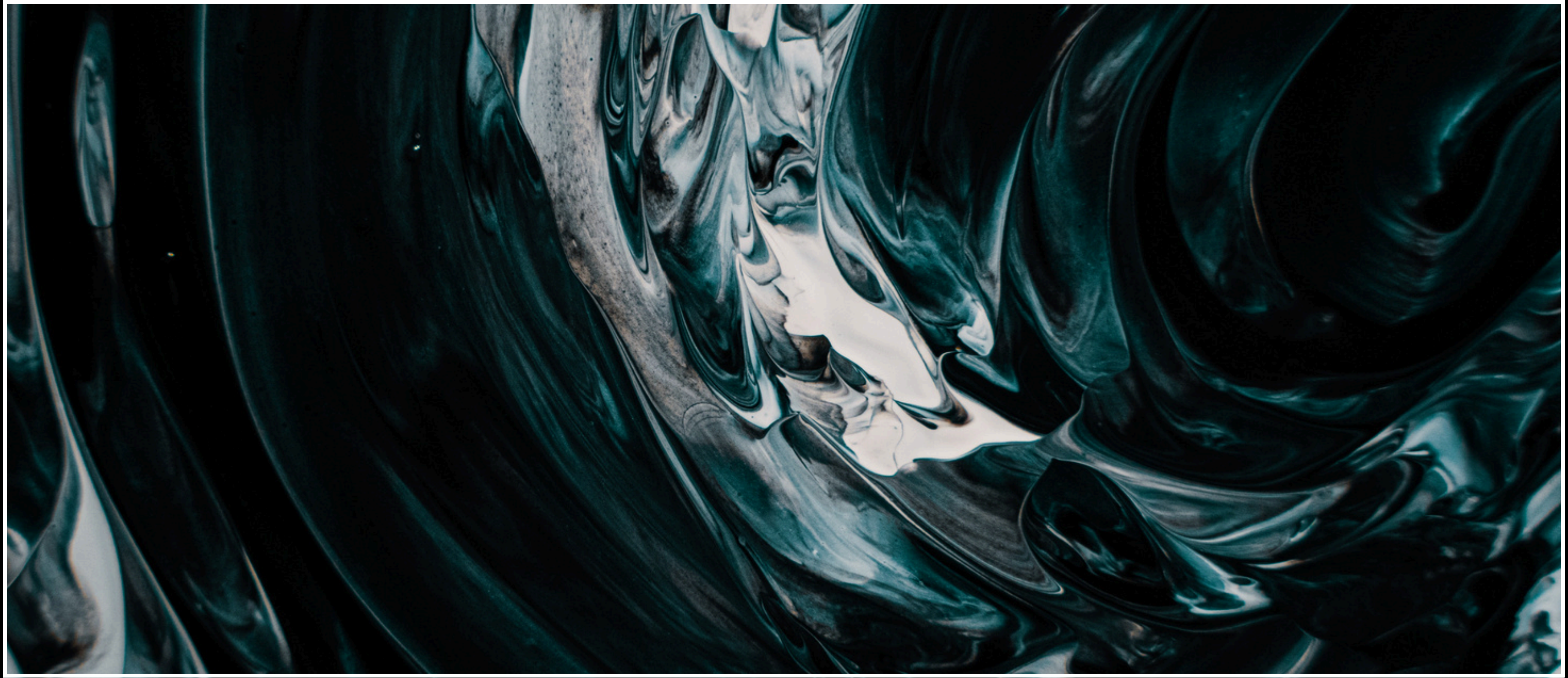
Key features

- Scaler-only Authentication: Google OAuth restricted to @sst.scaler.com — non-Scaler emails blocked.
- Role-based Access Control: Admin and user roles with protected routes and JWT sessions.
- Event Management: CRUD for multiple event types (Fest, Hackathon, Cultural, Townhall) with visibility controls and featured events.
- House System: Four houses (PHOENIX, TUSKER, LEO, KONG) with point awarding/deduction and transaction history.
- Real-time Leaderboard: Live standings & point history updates for immediate competition feedback.
- House-specific Registration: Single or per-house registration links for cultural events.
- Admin Dashboard: Create/edit events, manage users, award points and view analytics.
- Security & Hardening: Helmet, rate limiting, input validation, Mongo sanitization and CORS configured for the frontend.
- Developer-friendly: Vite + React frontend, Node/Express backend, MongoDB, documented API templates & seed scripts.

CASE STUDY (URBAN COMPANY)

Conducted a comprehensive business strategy analysis focusing on the real estate and society management sector. The goal was to identify scalability, retention, and revenue challenges faced by builders and propose a sustainable financial solution that aligns with both builder incentives and society needs.

Case Study Link :
https://docs.google.com/document/d/1f3hW2TAc8VsnFCNyycK5vNI6NNOHrO4gd_f7H0T2gF4/edit?usp=sharing



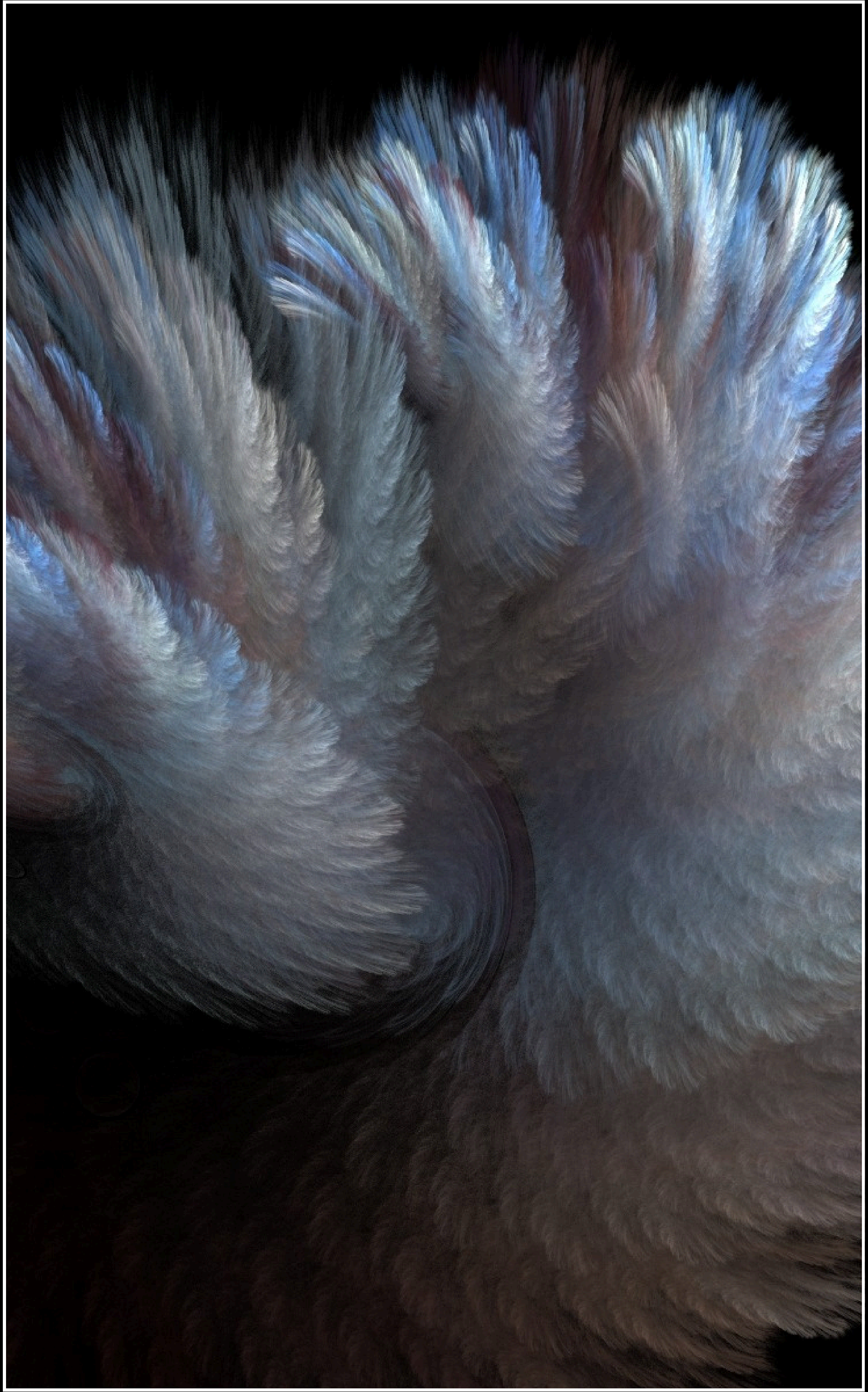
KEY CONTRIBUTIONS

- Market Positioning Analysis: Researched the competitive landscape to evaluate how existing service models for builders and society management operate.
- Gap Identification: Identified critical shortcomings in scalability, customer retention, and revenue generation within current practices.
- Competitive Benchmarking: Compared industry players on parameters like service offerings, customer adoption, and long-term sustainability.
- Field Research: Engaged directly with builders and society management teams to gather practical insights and validate assumptions.
- Proposed Financial Model: Designed a 5-year fixed-deposit (FD) service model offering builders a 10% benefit, with returns strategically allocated to society management for operational or welfare use.

AASRAH

Developed a web platform that connects users in distress with NGOs and volunteers, ensuring immediate help for people and animals in need. The platform acts as a bridge between individuals seeking support and NGOs capable of providing timely interventions.

GitHub Repo Link :
https://github.com/NoiceHax/Aasrah



KEY FEATURES

- Real-time Assistance: Users can raise alerts or requests for help (medical aid, animal rescue, disaster support, etc.), which are instantly visible to registered NGOs.
- Google Maps / Google Spatial Index (GSI) Integration: Implemented accurate location tracking and mapping, enabling NGOs to pinpoint the user's exact location and respond faster.
- Interactive Dashboard: NGOs and volunteers have a dashboard to view incoming requests, track their status, and assign team members.
- User-Friendly Interface: Designed an intuitive UI so users can quickly submit help requests with minimal effort during emergencies.
- Scalable Architecture: Built with scalability in mind to handle multiple NGOs, requests, and simultaneous users.

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WEBSITE

Portfolio Webpage

THANK YOU